



Welcome to the Qurified World

QR codes are those two dimensional grids of square pixels, featured prominently in some car advertisements. They look like tiles from some board game the Martians left behind. Today, they are being used to point mobile browsers to URLs, snap up contacts for the address books, record geo-location tags, or designate Wi-Fi hot spots. Creating your own QR code, or decoding one with your webcam, is very easy with no-fuss FOSS tools.

The QR or Quick Response code is a two-dimensional matrix code that is read with an image capture device rather than a linear scanner. Originally invented by the Toyota subsidiary Denso Wave to track the progress of automobiles in the manufacturing process about 20 years ago, it found its way into everyday use thanks to its ability to store more and virtually any kind of information compared to a linear bar code. The QR code is an ISO standard today. It is also licence-free. Though Denso Wave holds patent rights, it has chosen not to exercise them.

Creating and decoding QR codes

All of the following code creators work on Ubuntu 12.04 LTS with the latest updates applied. Your mileage may vary depending on your distribution.

Qreator: While Qreator can create QR codes as you type in your content, it cannot decode QR codes. It's an excellent first application to familiarise yourself with QR codes, and is available in the Ubuntu repositories. To get going, click the *New* button and then double-click one of the four options of *URL*, *Text*, *Geolocation*, or *WiFi Network*. Type in your URL or text, and see the square tiles of the QR code dance into place. Click the *Save* button on the bottom left to save the QR code as a PNG file.

Note a couple of things here: the more content you want the QR code to hold, the busier it gets. The busier the QR code, the less immune it is to mutation or distortion. We

have generated the '<http://www.linuxforu.com/>' QR code using Qreator, and taken the liberty to annotate it with the content string (using GIMP), but still managed to keep the code readable. The block-level error correction, even at its lowest level, makes the QR code resilient to some degree of distortion or erasure.

QtQr: If Qreator can create the code, QtQr can decode it, too. Just drag and drop the QR code image on the application window, and the pop-up window gives the decoded content of the QR code. If you have a printed QR code, you can get QtQr to activate your webcam to capture and decode the QR code. Hit *Ctrl+W* and a window identifying your webcam will appear, with brief instructions. After you confirm webcam activation, hold the printed code close to the camera, until a green square flashes around the code in the camera image. You will find that a window has already popped up with the content decoded.

At the command-line

The Linux experience is never complete without the command line, so... '*qrencode*' is a command-line tool to encode content to QR code. Read the man page, or use the '*--help*' option to figure out how to use it. The utility has quite a few capabilities, and some die-hard command line jockeys would say that it is more convenient. The only downside is that you have to provide your own encoding of the content. However, some might love the flexibility this offers. The